Descriptive Statistics

GENERAL ASSEMBLY

Learning Objectives

- Review pandas.
- Introduce the many ways statistics can help us learn about a dataset.
- Explain what commonly used summary statistics are and apply them to new datasets.
- Understand mathematical notation for formulas.

Descriptive statistics

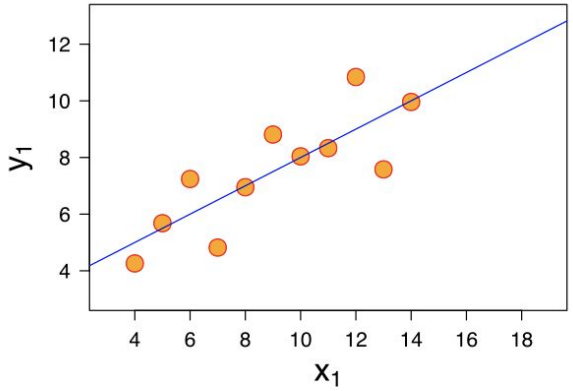
- We can use descriptive statistics to **summarize a dataset** using just a few numbers.
- This is often *really* important for communicating with other people about your work.
- Later in the course we'll also discuss **exploratory data analysis** (EDA) and **inferential statistics**.

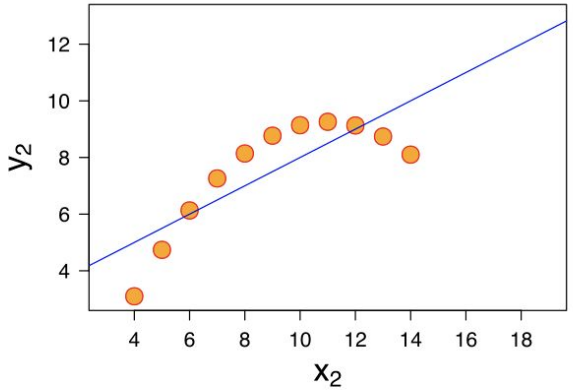
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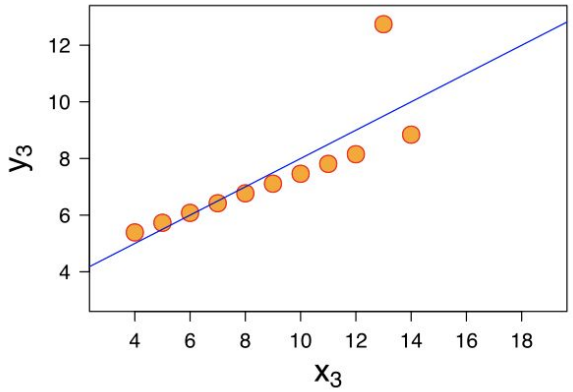
Caveat: descriptive statistics are *incredibly* useful, but we can't limit ourselves to them.

Each of these datasets has a mean of 9, a variance of 11, and a correlation score between x and y of .816.

Only when we **plot** these data do we see that they're *really* different.







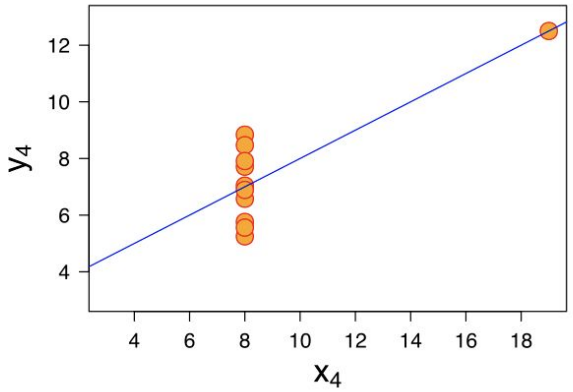


Image source: Wikipedia

Let’s look at some data!

Words to describe datasets

- Each row represents an **observation**.
- Each column represents a **variable**.
- Variables can be either **quantitative** or **categorical**.
  - Categorical variables correspond to **unordered, exclusive** categories.
  - Quantitative variables correspond to **numerical measures of magnitude or quantity**.
- The **number of observations** and the **number of variables** is important to know, too!

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Words to describe datasets

Continuous variables exist on an infinite spectrum and have infinitely many possible values. Continuous variables might end with decimals. Think about weights, prices, number of square feet, etc.

Discrete variables are usually counts. Think about number of children in a household, number of bedrooms in a house, etc.

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There are also **ordinal** variables that correspond to ranks on a scale. Think of customer satisfaction surveys after buying something online.

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Central tendency & spread

How to summarize data

When we start thinking about providing descriptive statistics, we'll need to consider what type of variable we're trying to describe.

When it comes to categorical variables, we're probably primarily interested in **counts**. When it comes to quantitative variables, we're interested in **central tendency** and **spread/variability**.

These help us communicate and describe datasets using just a few numbers.

Categorical summaries

How can we calculate the number of counts of each genre?

Genre counts could be presented in a table, or using a **bar plot**.

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Central tendency: mean/average

The average is usually the arithmetic mean. This is calculated by taking the sum of all data points, divided by the total number of points. The average is one way to ask, what's "typical"? What's "average"?

Central tendency: median

Calculated by sorting the data points from largest to smallest and identifying the number in the middle of the ordered list. “What’s in the middle?”

- The median is considered more **robust to outliers** than the mean.

Central tendency: mode

The number that shows up most often. Sometimes there's more than one mode, and sometimes the mode doesn't exist.

Variability: range

The biggest number minus the smallest number. How spread out is the data? Is it widely dispersed, or is it narrowly dispersed?

Variability: variance and standard deviation

The variance and standard deviation tell us about the typical distance from the mean.

Variability: variance and standard deviation

Variance: the average squared distance from the mean.

Variability: variance and standard deviation

Standard deviation: the square root of variance. Expressed in the same units as the data, making it easier to interpret.

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Covariance and correlation

(time permitting)

Covariance and correlation

- **Covariance** and **correlation** tell us about how two continuous variables are related.
- Correlation is more commonly reported than covariance because it's **normalized**. Covariance is used to calculate correlation, so we'll discuss both today, but we'll focus on correlation.

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Covariance and correlation

- Correlation: not causation
  - Spurious correlations
  - Correlation due to confounding
- “As *one variable* changes, what *tends to happen* to this other variable?”
- **Only measures linear relationships**; covariance and correlation scores do not well describe non-linear relationships between variables
  - “Linear”: like a line!

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Covariance and correlation: generally

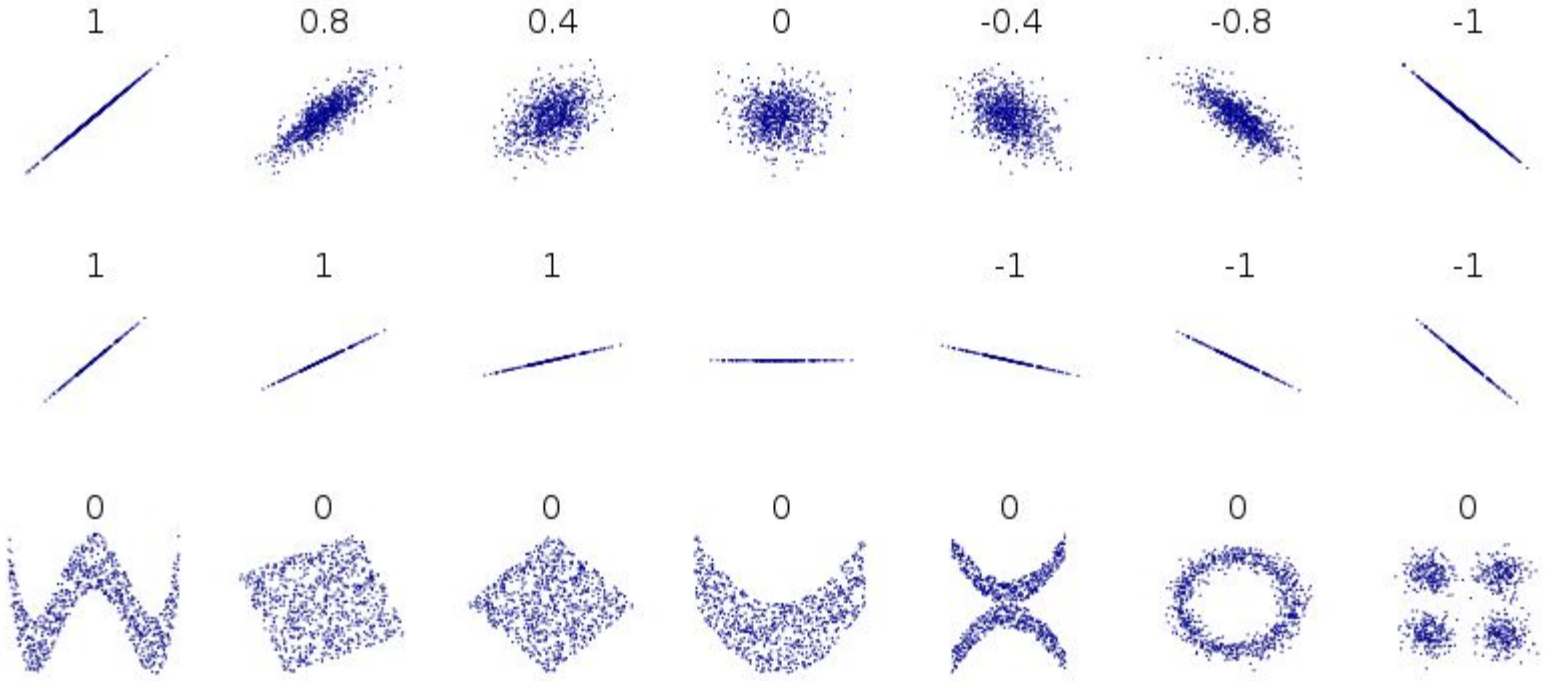


Image source: Wikipedia

Covariance and correlation: formula for covariance

Covariance and correlation: formula for correlation